

STAGE I · CONSTRUCT SCORING TARGETS



Inputs

Query q

Gold source $s(q)$



Scoring targets



Raw · $T_{\text{raw}}(q, S)$
only the exact raw source turn



Source · $T_{\text{src}}(q, S)$
any stored descendant of $s(q)$



Canonical · $T_{\text{can}}(q, S)$
transformed serving memories

$T_{\text{raw}} \subseteq T_{\text{src}} \supseteq T_{\text{can}}$
 $\text{Raw} \cap \text{Canonical} \approx \emptyset$ (usually disjoint)



STAGE II · RESCORE FIXED OUTPUTS



Saved outputs

Ranked trace $R(q)$

1 · m_7 · score 0.73
2 · m_3 · score 0.62
3 · m_9 · score 0.21



retrieval never rerun



Rescoring

T_{raw}

T_{src}

T_{can}

Shared-subset control

compare only queries that
all three targets cover

$$\Delta_t(A, B) = M_t(A) - M_t(B)$$

system gap under target t · $M = \text{nDCG}$



What can change

score shifts Δ_t

hit $\leftrightarrow$ miss flips

winner flips

design flips



Semantic audit · $n = 1,902$



fully supports $s(q)$ · only 29.2%



partial support / needs context



contradicts or irrelevant

Target noninvariance

benchmark conclusions can change
while ranked outputs stay fixed